

Ising model

G. Mallia

Main points

Lattices

Bonds

Spin Lattices

Boundaries

Hamiltonian

Spontaneous
Magnetisation

Critical
Temperature

Ising model

Giuseppe Mallia

g.mallia@imperial.ac.uk

Chemistry Department

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IMPERIAL

Outline

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- 3 Bonds
- 4 Spin Lattices
- 5 Boundaries
- 6 Hamiltonian
- 7 Spontaneous Magnetisation
- 8 Critical Temperature

Main points (1)

Ising model

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- Framework developed by Ernst Ising in the early 1920s; applied in various fields (i.e. neural networks).
"Spontaneous magnetization" was the subject of the PhD of Ising
- Ferromagnetism - Phase transition
- Local interaction / short-range interaction
- Global order / long-range correlative behaviour
- Small change in a parameter such as temperature
- Large-scale qualitative change in the state of a system

Main points (2)

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- Discrete variables (spins) arranged on a lattice, interacting via nearest-neighbor forces and external fields.
- Exact solutions in 1D and 2D
- In 1944, Lars Onsager solved the 2D Ising model (no field).
- Testing model for computational methods, in this specific lab for Monte Carlo simulations.

Lattices

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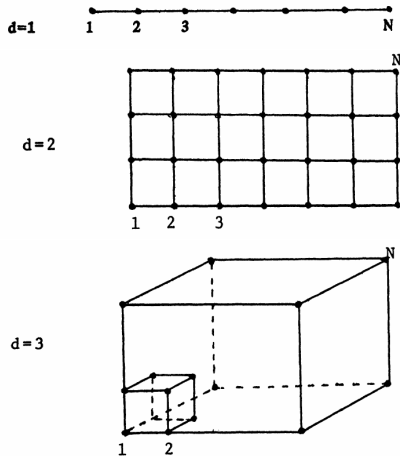
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Bonds

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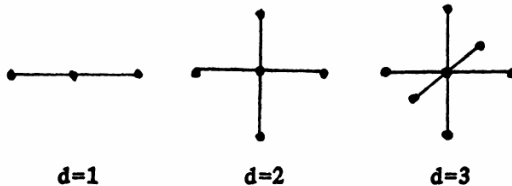
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Barry A. Cipra The American Mathematical Monthly, Vol. 94, No. 10. (Dec., 1987), pp. 937-959.

Spin Lattices

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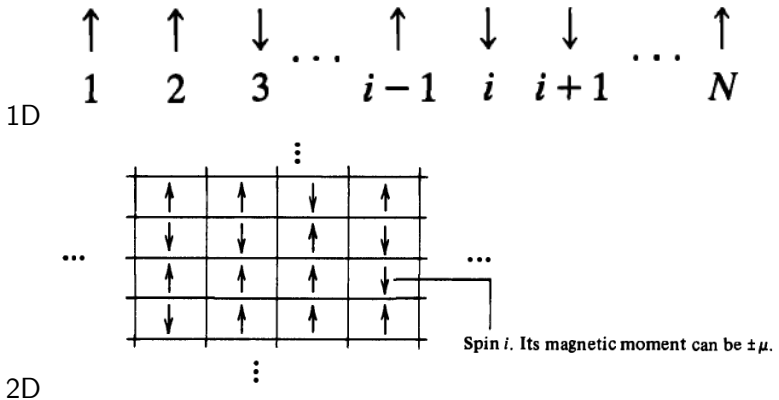
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Boundaries

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- Periodic Boundary Conditions
 - Symmetry
 - Each site has the same local environment/nearest neighbors
 - N sites are connected by dN bonds,
where d is the dimension of the lattice (1,2,3).
- Boundary Conditions
 - Direction of spontaneous magnetisation

Hamiltonian

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$$\hat{H} = -J \sum_{\langle i,j \rangle} s_i s_j - \sum_i h s_i$$

$$\hat{H} = -\frac{1}{2} J \sum_i \sum_{j_{nn}} s_i s_j - \sum_i h s_i$$

where j_{nn} nearest neighbors

s_i - the spin of a lattice site - can be either $+1$ or -1

h is the magnetic field

- $J > 0$: ferromagnetism
- $J < 0$: antiferromagnetism

In the generalised Ising model

s_i can assume various values ($-1, 0, 1$)

or a continuum of values

Spin interactions in the Ising model

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The spin system with the following nearest-neighbours spin interaction is called Ising model:

$$-J \sum_{\langle i,j \rangle} s_i s_j$$

$$-\frac{1}{2}J \sum_i \sum_{j_{nn}} s_i s_j$$

when $T = 0K$ and $J > 0$, in the most favourable configuration the spin will be aligned.

Spontaneous Magnetisation

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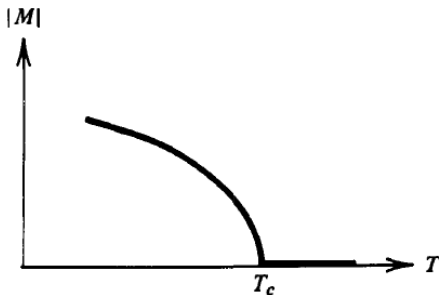
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In the absence of the external magnetic field the nearest neighbour spin alignment generates a cooperative long-range phenomenon: Spontaneous Magnetisation



Curie Temperature / Critical Temperature

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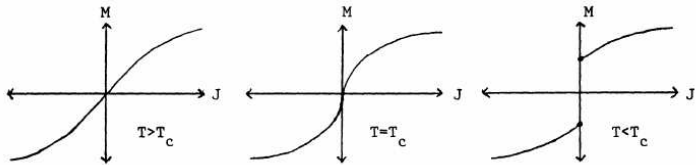
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The highest temperature at which the magnetisation is non-zero.

The system undergoes a phase transition / order-disorder transition.



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THANK YOU!!!

Questions